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Dynamic configurations and ecological fit: A fuzzy-set QCA analysis of effective data security governance in summer olympic games (1992–2024)

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ABSTRACT

Against the backdrop of accelerating digital transformation, Summer Olympic Games have become mega-temporary events generating massive volumes of data, where data security governance is critical to smooth event delivery. Nevertheless, existing scholarship mostly adopts fragmented technology-centric or management-centric perspectives, overlooking the synergistic interplay among multi-dimensional factors. Drawing upon information ecology theory, this study applies fuzzy-set Qualitative Comparative Analysis (fsQCA) to nine Summer Olympic Games held between 1992 and 2024. We examine how five antecedent conditions—data asset sensitivity, organizational management maturity, technological maturity, security deployment intensity, and external regulatory pressure—jointly shape high-level data security governance. Necessary-condition analysis shows that no single factor is indispensable for sound governance. Three equifinal configurational paths are identified: the Full-Element Driven path, the Management-Technology Dual-Driven path, and the Low-Risk Baseline path. Our findings demonstrate that effective Olympic data security governance arises from nonlinear multi-element synergy rather than isolated single-factor advantages, while governance failures stem from ecological mismatches among information-ecology components. Building on these patterns, we derive a path-contingent investment framework specifying what budget-constrained hosts can compress and what they must prioritize. This paper extends information ecology theory to the temporary governance context of mega-sporting events and delivers differentiated, resource-constrained managerial insights for future Olympic host cities under the ecological-fit principle.

1. Introduction

Against the backdrop of digital transformation, the Summer Olympic Games have evolved into one of the world's largest and most data-intensive mega-temporary events. From athlete biometric identification and real-time location tracking to concurrent streaming interactions across billions of terminal devices, each edition continuously sets new records in data generation and transmission. Simultaneously, this high-density data ecosystem has placed the Games at high risk of cyberattacks and data breaches; in 2012, London's power systems suffered massive cyber penetration; in 2016, Rio's World Anti-Doping Agency database was attacked, leading to the leakage of sensitive medical information of 41 athletes; during the 2020 Tokyo Games, approximately 450 million cyberattacks were recorded. These security incidents not only pose disruption threats to event operations but also easily trigger national reputational crises and global privacy litigation, compelling hosts to place data security governance at an unprecedented strategic priority. Yet, despite growing investment in technology and institutional development by organizing committees, data security incidents remain difficult to eradicate and have even trended toward greater complexity—the 2024 Paris Olympics, despite adopting security by design strategies and constructing a dual-layer defense architecture, still recorded 119 low-impact security incidents and 22 successful malicious intrusions during the Games. This paradox suggests that the effectiveness of Olympic data security governance is not determined by any single factor but rather emerges from the coupling of multiple forces within a complex environment.

Existing studies fall into three isolated streams: technical protection, organizational management, and regulatory compliance. Such fragmented perspectives implicitly assume linear additive effects: greater technical investment and stricter institutions would straightforwardly produce superior security outcomes. This linear view cannot explain real-world paradoxes: several Olympic editions with state-of-the-art cybersecurity resources still experienced severe data breaches.

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Furthermore, prior empirical work largely relies on single-event descriptive summaries, lacking systematic cross-edition comparative analysis of multi-factor synergies.

Information ecology theory provides a suitable holistic analytical lens. It conceptualizes information systems as organic ecosystems consisting of information, information people, information technology, and information environment, stressing dynamic co-adaptation rather than isolated element optimization (Nardi and O'Day, 1999). Despite its wide adoption in information-security research, this framework has rarely been deployed for mega-sporting event contexts.

To fill these gaps, this study addresses two progressive research questions:

RQ1. From an information-ecology perspective, are there any necessary conditions for high-level data-security governance of Summer Olympic Games?

RQ2. What sufficient condition configurations generate high-level data-security governance, and what equifinal paths exist across different Olympic editions?

This paper makes three key contributions. Theoretically, we extend information ecology theory from permanent enterprise information systems to the domain of mega-temporary event governance and advance configurational thinking for sport-event cybersecurity research. Methodologically, this work offers a replicable fsQCA analytical workflow for small-N comparative studies of event governance, with full transparency on calibration and robustness checks. Practically, instead of recommending universal high-input strategies, we propose context-sensitive ecological-fit governance alternatives for prospective host cities under budget and resource constraints.

The remainder of this paper proceeds as follows. Section 2 reviews prior literature and operationalizes our information-ecology-based conceptual framework. Section 3 describes case selection, data sources, and fuzzy-set calibration. Section 4 presents necessary-condition analysis and sufficiency configurational results. Section 5 discusses theoretical implications, practical managerial trade-offs, and study limitations. Section 6 concludes with future research directions.

2. Literature review

2.1. Research progress on data security governance in sports events

Data security governance in sports events is a research domain with distinctive interdisciplinary characteristics, situated at the intersection of information systems security, sports management, public policy, and legal regulation. In recent years, with the rapid advancement of digitalization in mega sports events, academic attention to this field has grown. Bongiovanni et al. (2024) conducted an integrative literature review that systematically mapped the academic landscape of cybersecurity research in sports, finding that although sport has become an important topic in management studies, academia has not yet sufficiently examined the role of cybersecurity and its strategic value in sports management, particularly event management (Bongiovanni et al., 2024). Their study classified existing literature based on the Technology-Organization-Environment (TOE) framework and explicitly identified sports event cybersecurity as "a seriously under-researched area that offers rich scientific opportunities." A research report from the University of California, Berkeley's Center for Long-Term Cybersecurity was even more pointed, noting that as mega sports events become increasingly digitalized, more and more segments—from scoring and refereeing systems to retail transactions, from spectator experiences to national security—are being incorporated into cyber-connected tracks, and new technologies bring both enormous opportunities and enormous risks. Furthermore, international academic discussions on security governance for major events have gradually shifted from single-dimension security assurance toward an integrative governance paradigm. The International Scientific Symposium on Security at Mega Sports Events held in Rabat in January 2026 explicitly advocated an integrative governance model based on foresight methods, operational coordination, and full-phase rigorous security planning, and encouraged scientific and applied research to bridge research findings with decision-making practice. These research dynamics collectively indicate that sports event data security governance is moving from a peripheral topic to the academic forefront, but overall theoretical construction remains at an early stage.

2.1.1. Technical protection research: from single-point defense to defense-in-depth systems

Technical protection is the dimension that has received the earliest systematic attention in data security governance research. Early studies focused on the deployment of encryption technologies and the application of intrusion detection systems. For example, Priebe et al. (2022) analyzed official application MY2022 for the 2022 Beijing Winter Olympics and found that the application contained encryption vulnerabilities, allowing user voice audio and file transfers to be easily bypassed, exposing weak links in mobile application security protection at mega events (Priebe et al., 2022). Meanwhile, the cyberattack landscape facing mega sports events has continued to escalate, with attacks exhibiting three major characteristics: targeted precision, industrialized teams, and massive scale, covering four attack surfaces: information systems, venue facilities, user groups, and supply chains (Wang et al., 2024). At the practical level, the national sports industry standard TY/T 1009.1–2024 Sports Events Informatization—Cybersecurity Specification, released in 2024, explicitly specifies the general cybersecurity architecture for sports events and cybersecurity work requirements during operations, marking the move toward standardization and institutionalization of sports event cybersecurity. However, despite the rich accumulation of technical protection research, its dominant paradigm remains centered on the superposition and optimization of "technical tools," tending to simplify security governance into a technical engineering problem, with limited theoretical attention to the interaction mechanisms among technology, organizational management, and the institutional environment.

2.1.2. Organizational management research: institutional construction and capacity building

Compared with the prolific technical protection research, academic research on the organizational management dimension of sports event data security governance remains relatively scare, yet they provide an indispensable institutional perspective for understanding governance effectiveness. In the big data era, China's digital transformation of mega sports events faces multiple governance challenges, including severe data silos and prominent data security issues. Wang (2025) categorized the obstacles to data governance in mega sports events into three types: foundational defects, systemic insufficiencies, and institutional weaknesses, and proposed strengthening governance foundations from the perspectives of multi-dimensional optimization systems, integrated platforms, and expanded application scenarios (Wang, 2025). This analytical framework reveals that the organizational management dilemmas of event data security governance cannot be resolved by technical investment alone, but require systematic institutional design and capacity building. At the international level, Dion-Schwarz et al. (2018) authored the Olympic-Caliber Cybersecurity report, systematically summarizing lessons learned from cybersecurity assurance across Olympic editions and providing a policy recommendation framework for the 2020 Tokyo Olympics and other major events (Dion-Schwarz et al., 2018). The core insight of this report is that Olympic-level cybersecurity is not merely a contest of technical capabilities, but a comprehensive manifestation of organizational coordination, cross-agency information sharing, and emergency response plan maturity. Overall, although the organizational management perspective reveals the importance of institutional capacity for data security governance, most studies remain at the level of single-dimension descriptive analysis and have not systematically examined technology,

management, and environmental elements within an integrative analytical framework.

2.1.3. Regulatory compliance research: institutional pressure and compliance dilemmas

Legal regulation constitutes the third major research dimension of sports event data security governance, and its importance has grown with the acceleration of global data protection legislation. The application of artificial intelligence technologies for security monitoring in mega sports events—such as intelligent video surveillance—has triggered widespread concerns around data privacy and algorithmic discrimination risks. Accordingly, regulatory debates over intelligent video surveillance in international mega-events such as the Olympics and FIFA World Cup have shifted from pre-event access approval to ongoing oversight of algorithmic decision-making processes. These discussions reflect the profound impact of global data protection regulations on event data governance. In particular, since the EU General Data Protection Regulation (GDPR) came into force in 2018, data processing activities involving European citizens' personal data at such events have faced unprecedented compliance pressure. Chinese scholars, in their research on data privacy protection and security management in sports events, treat legal regulation, ethical principles, and technical means as core analytical dimensions, exploring the role and limitations of laws and regulations in sports event data management, as well as the application and impact of ethical principles in data processing (Liu and Luo, 2025). At the practical level, the 2020 Tokyo Olympics faced dual compliance requirements of GDPR and Japan's Act on the Protection of Personal Information (APPI), and the Organizing Committee established a Data Protection Officer system specifically to ensure that data processing activities during event operations met multi-jurisdictional compliance standards. The 2022 Beijing Winter Olympics similarly set "meeting the data security and privacy protection regulations of nearly 200 countries, regions, and international organizations" as one of the core objectives of its security system design. However, existing research on regulatory compliance has a critical blind spot: it tends to frame legal regulation as a static external constraint, while overlooking its differentiated impacts on governance effectiveness across event phases. Specifically, institutional design pressures during the planning phase, emergency compliance requirements during the event, and post-event accountability mechanisms differ substantially in both nature and intensity.

2.1.4. Limitations of existing research and the entry point of this study

Synthesizing the literature across the three dimensions above, three structural limitations of existing research can be identified. First, **perspective fragmentation**. Technical protection research focuses on optimal technical solutions, organizational management research emphasizes institutional capacity building, and regulatory compliance research focuses on meeting compliance standards—the three operate in isolation, lacking an integrative perspective that places technology, organization, and environment under a unified theoretical framework for synergistic analysis. Second, **linear causal assumptions**. Existing research generally embeds an implicit assumption: the more technology is invested and the more complete the institutions and the stricter the regulations, the higher the governance effectiveness. However, the empirical facts of the Olympics repeatedly show that massive superpositions of technical and management resources do not always linearly translate into better governance outcomes; a "strong" single dimension may be constrained by a "weak" dimension in another. The effectiveness of data security governance is more likely a configurational effect of multi-element nonlinear synergy, rather than a net effect driven by single elements independently. Third, **insufficient systematic case comparison**. Existing studies are mostly single-event experience summaries or normative discussions, lacking systematic cross-edition, cross-era comparative analysis.

2.2. Information ecology theory and its applicability

2.2.1. Origins and core connotations of information ecology theory

Information Ecology is an interdisciplinary perspective that introduces the theoretical metaphor and methodology of natural ecology into information science. Its core tenet is that the operation of information systems is not a mechanical combination of isolated technical elements, but an organic ecosystem in which four categories of elements—information entities, information people, information technology, and information environment—are interdependent and co-evolve. The concept can be traced to the preliminary idea proposed by American scholar (Horton, 1978). In 1990, German scholar Rafael Capurro first explicitly coined the term "information ecology" (Capurro, 1990). The key figure who systematized this concept and pushed it to the forefront of academia and practice was Thomas Davenport, who in his 1997 book Information Ecology: Mastering the Information and Knowledge Environment formally treated information as an ecosystem and emphasized the importance of ecological metaphor for understanding organizational information management (Davenport, 1997). Almost simultaneously, anthropologists Nardi and O'Day (1999) conducted a deeper theoretical construction of information ecology from the perspective of human-computer interaction and socio-technical systems (Nardi and O'Day, 1999). They defined information ecology as a system constructed by people, practices, values, and technology in a given environment, and emphasized that its core focus "is not on technology, but on the human activities that technology serves." Nardi and O'Day further extracted five core analytical components from ecology: systemicity, diversity, co-evolution, keystone species, and locality. These components provide a solid operational foundation for subsequent empirical applications of information ecology. In their latest research in the field of digital welfare services, Pekkarinen et al. (2021) re-examined Nardi and O'Day's five elements of information ecology and found that in the digital age, these basic components still exist but need to be explained in new ways—such as the shift from closed systems to open systems, and the significantly enhanced meaning of digital diversity (Pekkarinen et al., 2021). This theoretical update indicates that information ecology theory is not a static classical framework, but remains a theoretically vibrant framework.

2.2.2. Existing applications of information ecology theory in information security governance

The application of information ecology theory in information security governance has achieved preliminary progress, providing important methodological references for this study. In Chinese scholarship, Qiu et al.(2022), from an information ecology perspective combined with modern governance theory, reconstructed a collaborative governance framework for national digital academic resource information security in cloud computing environments, systematically analyzing governance objects, governance subjects, governance environments, and the synergistic mechanisms within each element (Qiu and Hu, 2022). This study demonstrated the theoretical adaptability of information ecology in handling complex information security governance problems by mapping the four elements of information ecology onto the structural dimensions of the security governance system. In the industrial information service domain, data security governance research based on information ecology theory aims to strengthen data security, mine data value, and guarantee data services, constructing a data security governance framework for industrial information service platforms (Cao et al., 2024). Research in the field of network information content ecological security risk has further expanded the application boundaries of information ecology theory, proposing a theoretical framework centered on holistic governance and intelligent governance to address new security risks arising from the mutual penetration of information ecology elements (Zhou and Zhang, 2022). These studies collectively demonstrate that the core advantage of information ecology theory lies in its ability to incorporate heterogeneous elements—technology, people, institutions,

and environment—under a unified ecological metaphor for systematic analysis, avoiding the fragmentation trap of "seeing the trees but not the forest." However, to date, information ecology theory has not yet been introduced into the research domain of sports event data security governance, let alone used to analyze the dynamic governance of temporary mega-events with clear temporal cycles. This theoretical gap constitutes the key entry point of this study.

2.2.3. Applicability of information ecology theory to olympic data security governance

Applying information ecology theory to the analysis of Olympic data security governance is not a simple theoretical borrowing, but is based on a deep convergence between the governance object and the theoretical framework. This convergence is manifested at two levels.

First, **isomorphism of elements**. The Olympic data security governance system naturally contains all four categories of elements defined by information ecology theory. The massive athlete biometric data, spectator personal information, and event operational data involved in the Games form the core of the information element. The OCOG's information security teams, IOC data protection officers, third-party technology vendors, and even volunteers scattered across various positions constitute a diverse information people group. Encryption technologies, intrusion detection systems, threat intelligence platforms, and zero-trust architectures are typical manifestations of the information technology element. And GDPR and other global data protection regulations, the IOC's Data Governance Charter, the host country's cybersecurity legal system, and the increasingly complex global cyber threat landscape together constitute the multiple dimensions of the information environment. This high correspondence of element structure means that the analytical framework of information ecology theory can be directly applied to the analysis of Olympic data security governance without complex theoretical adaptation.

Second, **matching of logic**. The core challenge of Olympic data security governance lies not in the weakness of any single element—well-funded Olympic hosts typically have the ability to procure top-tier security technologies—but in the adaptation and synergy among multiple elements. Information ecology theory precisely emphasizes that system health does not depend on the optimization of any single element, but on the dynamic balance and organic interaction among elements. Nardi and O'Day explicitly stated that a healthy information ecology takes "people" rather than "technology" as its core, and technology can only play its expected role under appropriate institutional arrangements and effective human use (Nardi and O'Day, 1999). This insight is highly consistent with the common paradox in Olympic data security governance: why did some Olympics that invested heavily in deploying the most advanced security technologies of the time still fail to avoid major data security incidents? The answer is very likely hidden in the "ecological mismatch" among technology, management, institutions, and information asset characteristics.

2.3. Analytical framework: operationalizing information ecology theory

Based on information ecology theory and the specific context of Olympic data security governance, this study constructs the following analytical framework (Fig. 1):

Information Element: Data Asset Sensitivity. This measures the comprehensive risk exposure level of data processed by a given Olympic edition, encompassing three dimensions: first, the type and sensitivity of personal data, such as whether it involves sensitive data categories subject to special protection like biometric identification data, real-time location data, or health medical information. Second, the scale and density of data interaction, such as the number of downloads of the official event application and the transaction frequency of the online ticketing system. Third, the openness of data circulation, i.e., the sharing scope of event data among multiple stakeholders including the OCOG, International Sports Federations, National Olympic Committees, broadcasters, and sponsors.

Information People Element: Organizational Management Maturity. This measures the comprehensive level of the event organizer's data security management capability, covering the OCOG's security governance structure, personnel security literacy, emergency response procedures, and internal coordination mechanisms. This concept goes beyond a simple binary judgment of "whether there is a security management system" and instead focuses on the depth of the system and the effectiveness of execution—including the organizational hierarchy and decision-making authority of the information security team, the security access and audit mechanisms for third-party vendors, the privacy protection training coverage rate for all staff, and the emergency response time for major security incidents.

Information Technology Element: General Technology Maturity and Security Protection Deployment Intensity. General technology maturity measures the objective readiness level of the global information security technology environment in the year the Olympics are held—i.e., the toolkit available to the host, including the strength of mainstream encryption algorithms, the development stage of intrusion detection and threat intelligence technologies, and the availability of emerging security paradigms such as zero-trust architecture. This conditional variable reflects the external constraints from the technology supply side. Security protection deployment intensity measures the sufficiency and advancement of the security measures actually chosen and deployed by the given Olympic edition, including the depth of network perimeter protection, the coverage of endpoint security

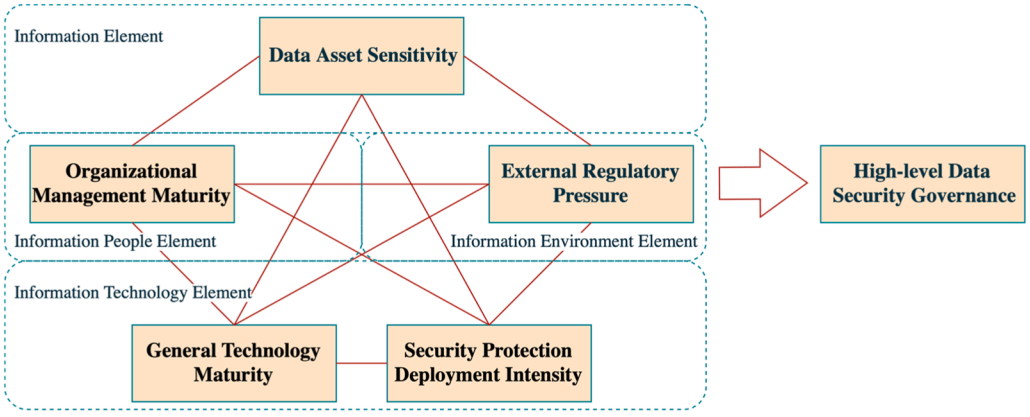


Fig. 1. Conceptual framework of olympic data security governance based on information ecology theory.

management, the granularity of data encryption, and the construction level of situational awareness platforms. The gap between the two reflects the moderating role of organizational choices and resource constraints on technology application.

Information Environment Element: External Regulatory Pressure. This measures the comprehensive level of data protection regulation strictness and international compliance requirements faced by the given Olympic edition, including the host country's personal information protection legislation status, the IOC's Data Governance Charter requirements, and cross-border data transfer legal restrictions.

2.4. Outcome variable: high-level data security governance

The outcome variable is **High-level Data Security Governance**, capturing real-world governance performance during each Summer Olympic Games' operational phase. Unlike most conditional variables which adopt positive calibration logic, this outcome uses reverse calibration anchored upon cybersecurity incident severity: better governance corresponds to fewer and less severe security breaches.

Incident evaluation adopts three explicit dimensions: (1) scope of impact (whether athletes, spectators, or core event operational systems are affected); (2) incident duration; (3) consequence severity (e.g., minor system disturbance, personal-data leakage, or large-scale operational disruption). We synthesize multi-source evidence including official post-event technical reports, third-party cybersecurity databases, peer-reviewed academic case analyses, and authoritative news archives to construct a composite performance score, which is further transformed into fuzzy-set membership scores.

Three qualitative anchors are set for outcome calibration:

·Full membership (0.95): No major security incidents, only negligible minor disturbances.

·Crossover point (0.50): Mixed outcomes with several moderate security events without systemic breakdown.

·Full non-membership (0.05): Large-scale data leakage or severe operational disruptions triggered by cybersecurity intrusions.

One critical measurement risk should be acknowledged: historical reporting bias. For early Olympic Games (1992-2004), cybersecurity attracted limited public attention, and host organizing committees had incentives to underreport minor security failures. To mitigate such bias, we avoid exclusive dependence on official host-committee documents and triangulate evidence across independent third-party archives. Even with multi-source cross-validation, complete elimination of under-reporting bias remains impossible, which constitutes an inherent measurement limitation of this retrospective study. The full evidence basis for each case's outcome membership score is documented in ***Supplementary Table S2***. The complete calibrated fuzzy-set membership matrix used for the main configurational analysis (first coder) is provided in ***Supplementary Table S3–1***; the independent second-coder ratings for reliability testing are provided in ***Supplementary Table S3–2***.

3. Research design

3.1. Methodological foundation: fsQCA

This study adopts fuzzy-set Qualitative Comparative Analysis (fsQCA) as the core research method. fsQCA was first systematically proposed by Ragin (1987) and is a case-oriented comparative research method based on Boolean algebra and fuzzy set theory (Ragin, 1987). Its core objective is to reveal the configurational paths through which condition combinations lead to outcomes by testing necessary conditions and sufficient conditions. The methodology is founded on three assumptions of causal complexity: concurrent causal relationships (multiple condition combinations jointly produce the outcome), equifinality (multiple different paths can lead to the same outcome), and causal asymmetry (the condition combinations for the occurrence and non-occurrence of the outcome are not mirror images).

The reasons for choosing fsQCA are threefold: First, the effectiveness of Olympic data security governance is the result of synergistic action of multiple factors, and traditional regression analysis methods struggle to effectively capture such multi-cause concurrent configurational effects. Second, the small-N sample of this study (9 Olympic editions) falls within the applicability range of fsQCA for case-oriented comparative research. The methodology typically recommends no >4–5 conditions for sample sizes above 10–15. Our sample size (N = 9) is close to this lower bound; therefore, we explicitly report the limited diversity issue and interpret the results with caution. Third, fsQCA can identify multiple equivalent paths leading to the same outcome, which is highly consistent with the "different paths, same destination" phenomenon in Olympic data security governance practice.

3.2. Case selection

This study selects all Summer Olympic Games from Barcelona 1992 to Paris 2024 as research cases, totaling nine editions. Case selection is based on the following considerations: First, Barcelona 1992 is widely regarded as the starting point of full-scale informatization of the modern Olympics, and since then the level of information technology application at each successive edition has continuously increased, forming an effective window for observing the evolution of data security governance. Second, the complete nine editions span over 30 years, covering the entire process of technological change from the pre-Internet era to the age of artificial intelligence and big data, presenting rich variation across all dimensions of information ecology elements. Third, the Summer Olympics have unparalleled benchmark significance in event scale, data ecosystem complexity, and global attention compared to other sports events, making theoretical construction based on these cases have the strongest external generalization potential.

This study focuses on the event-time operational period as a unified analytical cross-section. The event-time operational period is the peak of data interaction density and also the exposure period for real attacks; the final effectiveness of data security governance is most directly and intensively tested at this stage. Selecting the event-time operational period as the analytical cross-section maximally ensures comparability among cases and avoids introducing additional heterogeneity due to differences in governance logic across different preparation phases.

3.3. Data sources

The data sources of this study adopt a multi-source triangulation strategy, specifically including five categories: (1) Official Olympic technical reports, which systematically record the event's information technology architecture, security measures, and major incidents. (2) IOC documents and policy texts, providing institutional anchors for calibrating the external regulatory pressure condition. (3) Third-party cybersecurity incident databases and special reports, such as assessment reports from the University of California, Berkeley's Center for Long-Term Cybersecurity. (4) Records and analyses of information security incidents at each Olympic edition in academic research literature. (5) International regulatory texts and authoritative news reports, including the applicability of global data protection regulations such as GDPR at each Olympic edition, host country data protection legislation texts, and authoritative news reports on major cybersecurity incidents.

3.4. Fuzzy set calibration

Fuzzy set calibration is one of the most critical steps in fsQCA analysis, directly determining the membership scores of cases in condition sets. Ragin (2008) as well as Schneider and Wagemann (2012) systematically elaborated on the principles and methods of fuzzy set calibration. This study adopts the direct calibration method, setting three qualitative anchors for each condition and outcome variable: full membership (membership score=0.95), crossover point (membership

score=0.50, i.e., the maximum ambiguity point of the fuzzy set), and full non-membership (membership score=0.05). The calibration anchors are set based on theoretical thresholds, key historical turning points, and substantive differences among cases.

Five conditional variables are derived from information-ecology dimensions: data asset sensitivity (DASENS), organizational management maturity (ORGMAT), general technology maturity (TECHMAT), security protection deployment intensity (SECDEP), and external regulatory pressure (REGPRESS). Each condition is decomposed into multiple observable sub-dimensions. Brief illustrative case examples for each anchor are provided below; exhaustive scoring rubrics are available in ***Supplementary Table S1***.

For early editions (1992-2004), publicly available cybersecurity documentation is sparse. Therefore, membership assignment strictly follows triangulation rules: at least two independent evidence sources (official technical reports, IOC archival materials, contemporary industry reports, or peer-reviewed historical studies) are required before assigning final membership values. All supporting references for each case are listed in ***Supplementary Table S2***.

To assess the reliability of the calibration, an independent second coder — blind to the original membership scores — re-coded all 54 case-condition judgments (9 editions × 6 variables) following the scoring rubric in ***Supplementary Table S1*** and the evidence base in ***Supplementary Table S2***. Because the substantive question for fsQCA is whether coding decisions alter set membership rather than whether two coders reproduce identical decimals, we report Krippendorff's α at two levels. At the nominal level based on calibration zones (full non-membership 〈 0.50, crossover = 0.50, full membership 〉 0.50), $\alpha = 0.93$, with 52 of 54 judgments (96.3%) falling in the same calibration zone. At the interval level of raw fuzzy scores, $\alpha = 0.99$, and 81.5% of judgments agreed within a tolerance of ±0.05.

The two zone-level disagreements both concern Rio 2016. The ORGMAT crossing (0.55 vs. 0.48) involves precisely the variable identified as the core failure mechanism in the counter-case analysis — the second coder's lower score reinforces, rather than contradicts, the interpretation of insufficient organizational-management maturity. The REGPRESS judgment (0.50 vs. 0.55) concerns a condition on which Rio sits at the point of maximum ambiguity in the original calibration. No disagreement altered any case's set membership on the outcome variable.

Full reliability statistics and details of disagreements are reported in ***Supplementary Table S5***; full paired membership scores for both coders are provided in ***Supplementary Tables S3–1*** and ***S3–2***. As a complementary robustness check, we additionally perturbed the crossover threshold (0.50 ± 0.05) and re-ran the entire analytical workflow; the results, reported in ***Supplementary Table S4*** and Section 4.4, confirm that the configurational structure is identical across all crossover-point variants (overall consistency = 0.962; overall coverage = 0.811).

Calibration of Conditional Variables

Data Asset Sensitivity (DASENS). The full membership anchor corresponds to editions with highly complex data ecosystems (e.g., Paris 2024, which employed AI-driven smart camera control and biometric identification). The full non-membership anchor corresponds to editions with only basic information systems and limited-scale personal information processing. The crossover point corresponds to intermediate editions where the data ecosystem underwent significant qualitative change.

Organizational Management Maturity (ORGMAT). Based on the assessment framework of the ISO/IEC 27,001 Information Security Management System standard, a comprehensive score is assigned to the process completeness described in public Olympic official reports and academic literature. The full membership anchor corresponds to editions that performed excellently across all dimensions of information security governance structure, personnel training, vendor auditing, and emergency response. The full non-membership anchor corresponds to early editions with obvious gaps in multiple key dimensions. The crossover point takes the intermediate level.

General Technology Maturity (TECHMAT). Calibrated based on the objective readiness level of the global information security technology environment in the year the Olympics were held. Full non-membership corresponds to the early stage when encryption technology was not yet mature. The crossover point roughly corresponds to the maturity node of threat intelligence and defense-in-depth systems in the mid-2000s. Full membership corresponds to the latest stage where zero-trust and AI-driven security have entered mainstream practice.

Security Protection Deployment Intensity (SECDEP). Measures the sufficiency of security measures actually deployed by the given Olympic edition relative to the technologies available in the same period. The full membership anchor corresponds to editions that implemented advanced protection across all layers of defense-in-depth, endpoint management, data encryption, and situational awareness. The full non-membership anchor corresponds to the earliest editions with the most basic protection deployment.

External Regulatory Pressure (REGPRESS). Takes the official implementation of GDPR in 2018 as the core institutional watershed. The full membership anchor corresponds to editions after GDPR took effect. The full non-membership anchor corresponds to the 1990s when global data protection regulations had not yet taken shape. The crossover point corresponds to the significant inflection point of rising institutional pressure.

Calibration of the Outcome Variable

The calibration of **High-level Data Security Governance (OUTCOME)** adopts a reverse calibration logic based on event severity. The higher the governance level, the closer the membership score is to 1, meaning the fewer and less severe the data security incidents at that Olympic edition. Full membership corresponds to editions with no major security incidents and very few minor incidents. Full non-membership corresponds to editions with multiple types of large-scale data security incidents. The crossover point corresponds to the intermediate level.

It should be noted that as a study based on secondary data, the precision of condition calibration is constrained by the incompleteness of public information and differences in reporting standards across editions. To mitigate this limitation, this study adopts the following strategies: (1) Prioritizing reliance on official technical reports and peer-reviewed academic literature in calibration, anchoring calibration points with objective indicators as much as possible; (2) Conducting robustness checks after empirical analysis, including adjusting calibration anchors and changing crossover point positions, to assess the sensitivity of core conclusions to calibration decisions. The complete calibrated fuzzy-set membership matrix used for the main configurational analysis (first coder) is provided in ***Supplementary Table S3–1***; the independent second-coder ratings for reliability testing are provided in ***Supplementary Table S3–2***.

4. Data analysis and empirical results

4.1. Necessary condition analysis

Before conducting sufficiency configurational analysis, we first examine whether any single condition constitutes a necessary condition for high-level data security governance. Following conventional QCA methodology standards, when a condition (or its negation) reaches a consistency of 0.90, it can be considered a necessary condition (Ragin, 2008).

Table 1 reports the necessary condition analysis results for each condition. The consistency of all conditions and their negations is below the 0.90 threshold, with the highest being non-high regulatory pressure (~REGPRESS) at a consistency of 0.729 and coverage of 0.973, but still not reaching the necessary condition standard. This indicates that within the information ecology framework, no single element can independently constitute a bottleneck for high-level data security governance. The improvement of governance performance does not depend on the

Table 1
Necessary condition analysis results.

Condition	Consistency	Coverage	Condition	Consistency	Coverage
DASENS	0.499	0.894	~DASENS	0.650	1.000
ORGMAT	0.628	0.951	~ORGMAT	0.534	0.975
TECHMAT	0.572	0.895	~TECHMAT	0.569	1.000
SECDEP	0.587	0.922	~SECDEP	0.572	1.000
REGPRESS	0.430	0.922	~REGPRESS	0.729	0.973

Note: No single condition or its negation reaches the 0.90 consistency threshold for necessary conditions. The coverage values for ~DASENS, ~TECHMAT, and ~SECDEP each equal 1.000, indicating that all cases with low levels of these respective conditions achieve high governance performance, with no counterexamples.

extreme strengthening of any single dimension, but rather on the configurational effect of multi-element synergistic adaptation. This finding preliminarily confirms the core proposition of information ecology theory—system health depends on the dynamic balance among elements, not on the optimization of a single element.

4.2. Sufficiency analysis

We perform truth-table sufficiency analysis with consistency threshold set to 0.80 and frequency threshold equal to 1. Due to our small-N sample (N = 9), limited diversity exists within the truth-table configurations. Following conventional QCA practice (Ragin, 2008), we attempted Boolean minimization including logical remainders; however, even after incorporating all logical remainders, no substantively meaningful parsimonious or intermediate solutions emerge. Consequently, only the complex solution is reported, and we refrain from distinguishing core versus peripheral conditions. Readers should interpret identified paths as exploratory equifinal configurations rather than deterministic causal laws. Four truth-table rows are included for positive-outcome analysis, covering eight high-governance cases, whereas Rio 2016 (membership score = 0.35) is excluded, consistent with its real-world governance failure. Boolean minimization yields three equifinal configurational paths for high-level data-security governance (see Table 2). The solution overall consistency reaches 0.962 and overall coverage equals 0.811.

We focus our primary analysis on configurations yielding high-level governance (OUTCOME). We also attempted truth-table analysis for the negated outcome (~OUTCOME, low-level governance). However, Rio 2016 is virtually the sole representative case of low governance performance in our sample. Such an extremely small number of negative cases cannot generate stable, interpretable configurational solutions for failure. Instead of reporting unreliable ~OUTCOME configurations, we conduct an in-depth single-case counter-factual discussion of Rio 2016 in Section 4.3. Future research collecting additional failure cases could revisit configurational analysis of low-governance outcomes.

P1: Full-Element Driven (DASENS·ORGMAT·TECHMAT·SECDEP). This path is conditioned on high data asset sensitivity, high organizational management maturity, high general technology maturity, and high security protection deployment intensity, with external regulatory pressure as an irrelevant condition. Its core logic is that when the data ecosystem is highly complex and the information load is heavy, full-element synergy among organizational management, the technology environment, and security deployment must be achieved; a shortfall in any single dimension may lead to governance failure. This path covers three contemporary Olympics: London 2012, Tokyo 2020, and Paris 2024. These three editions all faced high-density data interaction and global cyber threats—London suffered power system penetration, Tokyo recorded 450 million cyberattacks, and Paris was projected to face an attack volume 8 to 10 times that of Tokyo—but all successfully maintained data security through the dual guarantee of management maturity and advanced technology. Notably, external regulatory pressure is an irrelevant condition in P1, meaning that when all elements are in place, the governance system's dependence on the institutional environment is relatively low, and internal management and technology capabilities can serve as the dominant forces.

P2: Management-Technology Dual-Driven (ORGMAT·TECHMAT·SECDEP·~REGPRESS). This path takes high organizational management maturity, high general technology maturity, and high security protection deployment intensity as core conditions, while external regulatory pressure is explicitly low (i.e., absent condition). Its logical connotation is that in an external environment where institutional regulation is not yet complete, the event host can achieve high-level governance by independently constructing a management system and deploying advanced technology. This path covers two Olympic editions: Beijing 2008 and London 2012. Beijing 2008 was situated in an era when global data protection regulations were not yet well-developed, but it established a solid organizational management and technology defense line by conducting risk assessments and systematic rectification with reference to the ISO 27,001 standard. Although London 2012 had begun to face rising institutional pressure, the overall governance effectiveness during event operations was still primarily driven by internal management maturity and technology deployment.

Path P3: Low-Risk Baseline (~DASENS·~ORGMAT·~TECHMAT·~SECDEP·~REGPRESS). This path is characterized by low levels of all five conditional variables. Its logical connotation is that when the sensitivity and interaction scale of data assets are at an extremely low level, the information load borne by the governance system is limited, and even if other ecological elements are not yet mature, high-level data security governance can still be achieved. This path covers four early Olympic editions from Barcelona 1992 to Athens 2004. The digitalization level of these events was relatively low—Barcelona involved only basic information systems and proprietary networks, Atlanta had just introduced an official website, and the online interactions of Sydney and Athens were far from reaching the scale of the contemporary era—so the probability and scope of data security incidents were naturally limited. Therefore, their governance success is mainly attributable to the low-risk exposure brought by low digitalization, rather than outstanding governance capability. Conversely, this path also demonstrates necessity of paths P1 and P2: as data sensitivity continues to rise, the low-risk baseline path will gradually exit the historical stage, and hosts must transition toward the full-element driven or management-technology dual-driven paths.

Table 2
Configurational paths to high-level data security governance.

Condition	Path1 (P1)	Path2 (P2)	Path3 (P3)
DASENS	●	-	◦
ORGMAT	●	●	◦
TECHMAT	●	●	◦
SECDEP	●	●	◦
REGPRESS	-	◦	◦
Consistency	0.939	0.945	1.000
Raw Coverage	0.499	0.344	0.513
Unique Coverage	0.209	0.017	0.258
Overall Consistency	0.962		
Overall Coverage	0.811		

Note: ● = condition present; ◦ = condition absent; - = condition irrelevant (does not affect the outcome).

4.3. Cross-Path comparison and theoretical dialogue

The three paths present a clear gradient of Olympic data security governance configurations, evolving from "low-risk baseline" to "management-technology synergy" and then to "full-element integration"(see Fig. 2). This gradient is not a simple temporal evolution—Beijing 2008 and London 2012 differ significantly in regulatory environment but share the core logic of P2—but rather reflects the functional substitution and contextual adaptation among information ecology elements. Under low information load (P3), the ecosystem can maintain a steady state even when all elements are at low levels, which maps the ecological principle of "wide-range equilibrium under low carrying pressure" onto the governance domain. When information load exceeds a certain threshold, the information people element and information technology elements first form a synergy, constituting the "management-technology dual-driven" steady state of P2, which explicitly requires low external regulatory pressure as a boundary condition. When information load escalates further to an extreme level, the four core elements—data assets, organization, technology, and security deployment—must all reach high levels to form the "full-element high-pressure equilibrium" of P1. Notably, external regulatory pressure becomes an irrelevant condition in P1: when internal capabilities are comprehensively strong, high governance performance can be achieved regardless of whether external regulatory pressure is high or low. This reveals a threshold effect of the information environment element in the ecology: it acts as a binding constraint when system capacity is insufficient, but its influence is diluted when internal elements are fully developed.

The failure of Rio 2016 provides powerful counter-evidence for the above theoretical interpretation. During Rio's event-time operational period, data asset sensitivity had reached 0.72, belonging to a high information load context, but its organizational management maturity was only 0.55, significantly lower than that of contemporary high-governance Olympic editions. According to the logic of information ecology theory, Rio was in a state of ecological imbalance caused by "high information load" requiring "high management adaptation" but actual management lagging behind. The direct consequence of this imbalance was the World Anti-Doping Agency (WADA) database attack that led to the leakage of sensitive medical information of 41 athletes. Rio's lesson reinforces a core insight of information ecology theory: system health does not depend on the strength of any single element, but on the adaptation among multiple elements. When the data sensitivity information element has already surged, the information people and information technology elements must upgrade synchronously, otherwise ecological imbalance and governance failure are inevitable.

4.4. Robustness test

We conduct multiple robustness exercises to evaluate configurational stability: (1) adjusting the consistency threshold to 0.75 and 0.85; (2) perturbing the calibration crossover point by ±0.05; and (3) iterative case-dropping tests. In addition to the inter-coder reliability assessment reported in Section 3.4 and ***Supplementary Table S5***, these exercises jointly assess whether the findings are artefacts of specific analytical parameters. Full outputs are compiled in ***Supplementary Table S4***.

Threshold-based checks show a stable core structure. When the consistency threshold is relaxed to 0.75, the three-path structure remains fully intact. Perturbing the crossover point to 0.45 and 0.55 leaves the configurational structure identical, with the overall consistency unchanged at 0.962, the overall coverage unchanged at 0.811, and no condition switching between core and peripheral status. Raising the consistency threshold to 0.85 merges Path 1 and Path 2 into a broader configuration — a consistent superset of the two separate paths — without reversing the direction of any core condition.

Case-dropping tests delimit the scope of two configurations. Path 1 and Path 3 remain highly stable across single-case deletions, whereas Path 2 disappears when Beijing 2008 is removed and Path 3 disappears when Barcelona 1992 is removed, indicating that these two lower-complexity paths are case-bound to their respective anchor cases. Both patterns are consistent with the exploratory character of small-N fsQCA designs.

Taken together, the robustness checks confirm that the core configurational findings are stable across threshold and calibration adjustments, while the case-dependence of individual paths is consistent with the small sample size of this exploratory analysis.

5. Discussion

5.1. Theoretical implications

This study delivers several theoretical insights for mega-event cybersecurity and information-ecology research.

First, we extend information ecology theory into the empirical setting of mega-temporary sporting events. Prior information-ecology

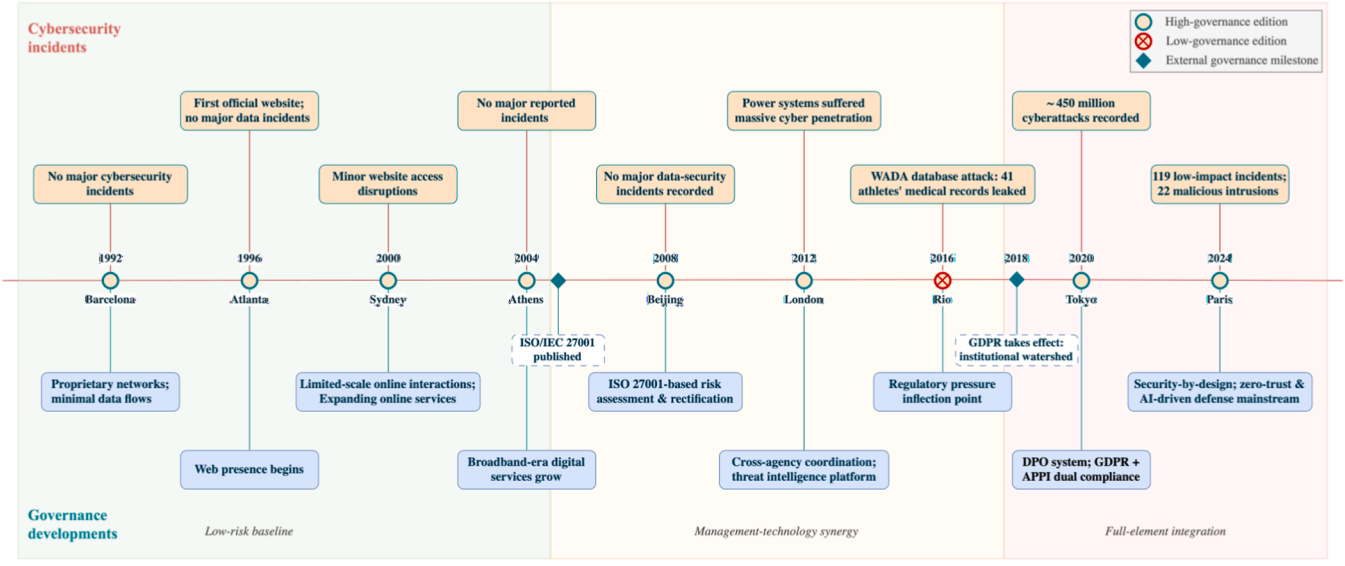


Fig. 2. Timeline of major cybersecurity incidents and data-security governance developments across the Summer Olympic Games (1992–2024).

applications mainly focused on persistent enterprise or public-sector information systems. Olympic Games represent short-lived, time-bounded mega-temporary ecosystems with rapid build-up and rapid dismantling. Our empirical results validate that the core ecological premise — system health depends on multi-element dynamic adaptation instead of single-element supremacy — also holds for temporary event contexts. The concept of "ecological mismatch" helps interpret governance failure such as Rio 2016, where surging data-asset sensitivity was not matched by synchronized organizational-management upgrading.

Second, our work contributes to a paradigm shift for sport-event cybersecurity research. Existing literature implicitly adopts linear net-effect thinking, expecting that adding more technical or institutional resources will linearly boost security performance. By applying fsQCA, this study demonstrates equifinality and concurrent causation: multiple distinct combinations can produce equally high-quality governance outcomes. Therefore, researchers should move beyond isolated factor regression toward configurational, synergistic analytical perspectives for mega-event security.

Third, this paper adds empirical evidence for cybersecurity governance contingency thinking. No universal one-size-fits-all optimal configuration exists. Governance effectiveness hinges on contextual ecological niches defined by data-asset features, organizational capacity, technological readiness and external regulatory environments.

5.2. Practical implications: managerial trade-offs under resource constraints

Superficially, the Full-Element Driven path (P1) could appear self-evident: that high data sensitivity requires high capabilities across technology, security deployment and management. However, the equifinal configurational results reveal four counter-intuitive managerial trade-offs that challenge the default assumption that "more of every dimension is always better", particularly for host cities operating under severe budget constraints.

First, prioritize organizational investment over technical hardware in procurement decisions. The default priority for mega-event security procurement is cutting-edge technical hardware. However, under low external regulatory pressure (the P2 context), mature organizational management capacity can partially compensate for relative disadvantages in technological maturity. For budget-constrained hosts, investing first in vendor auditing systems, cross-agency incident-response drills and organisation-wide security training delivers higher governance returns than blindly upgrading cybersecurity hardware. This runs counter to common procurement logic and aligns with the core insight of the Management-Technology Dual-Driven path: management capacity, not technology, acts as the foundational driver.

Second, regulatory pressure is not a substitute for internal capability. Conventional wisdom holds that stricter data protection regulation automatically improves security outcomes, and that compliance investment is therefore universally the highest priority. However, our configurational results indicate that once internal management, technology and security deployment are simultaneously robust, high governance performance is achieved irrespective of the level of external regulatory pressure. Regulatory pressure operates as a binding constraint only when internal system capacity is insufficient, and its influence is diluted once internal elements are fully developed. The managerial implication is therefore a matter of sequencing rather than absolute spending: compliance investment alone, without commensurate internal capability, does not deliver high-level governance. We emphazise that this finding should not be read as a recommendation to defer regulatory compliance — all three P1 cases operated under non-trivial regulatory pressure, and the post-GDPR trajectory means compliance-capable technology is increasingly a baseline requirement rather than a discretionary item.

Third, downsizing data scope is a parsimonious alternative to full-element escalation. The default trajectory for Olympic host cities is continuous expansion of digital services and data collection, assuming greater digitalization always delivers better event experiences. However, host cities can deliberately adjust their information-ecology niche by constraining the scope of high-risk sensitive data (e.g., reducing non-essential biometric data collection, limiting third-party data sharing, shortening retention windows). Where operationally acceptable, this can shift a Games from the P1 regime into the P2 or even P3 range, allowing acceptable security to be achieved with far lower resource input. This "ecological niche adjustment" strategy provides a deliberate alternative to the default logic of ever-increasing digitalization.

Fourth, organizational gaps are more dangerous than technical gaps in incident response. When security incidents do occur, the default policy response is to increase technical investment. However, the Rio 2016 counter-example demonstrates that rising data sensitivity cannot be offset by technical investment alone if organizational management maturity lags behind. Under budget constraints, closing organizational capacity gaps should take priority over upgrading technical tools — the opposite of common procurement priorities for mega-events, and a sequencing rule that follows directly from the failure mechanism identified in the counter-case analysis.

Together, these trade-offs demonstrate that the contribution of the configurational analysis lies not in stating that high-complexity Games need high capabilities, but in identifying which dimensions can be substituted for others, which strategies can reduce resource requirements without compromising security, and when conventional wisdom about security investment does not hold.

5.3. Limitations of the present study

Several important limitations should be acknowledged.

First, sample constraints. We analyze only nine Summer Olympic editions. The small-N sample falls near the lower bound of fsQCA applicability. Our configurations are exploratory findings; generalization to Winter Olympic Games or other mega-events requires caution. Limited diversity prevents the reporting of parsimonious solutions and distinguishing core/peripheral conditions.

Second, measurement constraints from retrospective secondary-data sources. Despite multi-source triangulation, historical under-reporting of cybersecurity incidents cannot be fully eliminated. Calibration anchors rely partly on substantive expert inference for pre-2000 editions, and although an independent second coder reproduced the calibration with high reliability (Krippendorff's $\alpha = 0.93$ at the calibration-zone level; ***Supplementary Table S5***), both codings drew on the same body of public archival records. Systematic biases shared by both coders — for example, reliance on under-reported cybersecurity incident data for pre-2000 editions — therefore cannot be fully ruled out. We mitigate this risk via crossover-point sensitivity robustness checks (Section 4.4, ***Supplementary Table S4***).

Third, variable granularity. Five high-level aggregated conditional variables are adopted in this paper. Each condition contains multiple sub-dimensions whose independent internal effects cannot be unpacked in the current analytical setup.

Fourth, this study focuses exclusively on in-event operational periods. Pre-game preparation and post-event legacy phases fall outside our analytical cross-section, and governance dynamics in these phases may differ substantially.

6. Conclusions and future research

6.1. Conclusions

This paper adopts information ecology theory and fsQCA to investigate high-level data-security governance across nine Summer Olympic Games from 1992-2024. Four major conclusions are reached.

First, no single condition constitutes a necessary requirement for

successful data-security governance for Summer Olympics. Good governance is a configurational outcome of multi-element synergy.

Second, three equifinal paths can yield high-level governance: Full-Element Driven, Management-Technology Dual-Driven, and Low-Risk Baseline.

Third, security-governance failures stem from ecological mismatch among information-ecology components, as demonstrated by the Rio 2016 counter-case. When data-asset sensitivity surges without synchronized upgrading of organizational-management or technical protection capacities, major security incidents are highly likely.

Fourth, host cities should adopt ecological-fit strategic thinking rather than blind resource-maximization, especially when operating under tight budget constraints.

6.2. *Future research directions*

Several promising avenues open for further research.

First, enlarge case repositories. Future studies may integrate Winter Olympic Games, World-Cup tournaments and other mega sporting events to build larger-N datasets. More low-performance failure cases would enable configurational analysis of negative governance outcomes, which our current sample cannot reliably support.

Second, pursue finer-grained measurement. Decompose our aggregated five conditions into multi-dimensional sub-indicators and apply multi-level QCA designs to unpack internal sub-component effects.

Third, combine set-theoretic configurational analysis with process-tracing qualitative methods. Within-case process tracing could reveal fine-grained micro-causal mechanisms behind each Olympic edition's security successes and failures, complementing cross-case fsQCA results.

Fourth, develop specialized measurement frameworks for temporary-event data-security governance, tailored for short-term mega-event contexts instead of re-using enterprise-oriented security metrics.

Fifth, comparative institutional analysis can examine how national-level governance quality and cybersecurity readiness moderate the compensation/substitution relationships among information-ecology elements across different host-country contexts.

Despite the above limitations, as the first exploration of introducing information ecology theory and set-theoretic methods into Olympic data security governance, this study provides hosts with a governance perspective based on ecological fit, and paves the way for a paradigm shift in academic research from "single-element influence" to "multi-element configuration". As digitalization deepens in future Olympic Games, the theoretical value and practical significance of this perspective will only increase.

CRediT authorship contribution statement

Xiaoyu Li: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Xinyan Guo:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiaoyu Li reports financial support was provided by Sichuan Office of Philosophy and Social Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cose.2026.105219.

Data availability

Research Link Provided
Olympic_qca_cross_section (Original data) (Mendeley Data)

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